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Standard compliance bounds origin–destination identifiability in GBFS bike-sharing feeds: a cost-learning and sampling-horizon analysis

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Keywords: GBFS; origin–destination reconstruction; data-standard compliance; entropic optimal transport; shared micromobility**Abstract**

Reconstructing origin–destination (OD) flows from public bike-sharing feeds is split between two regimes: macroscopic inference from station stock counts, which is underdetermined, and deterministic tracking of per-vehicle identifiers, which is exposed to temporal aliasing and to the privacy-driven rotation of identifiers recommended by the GBFS standard. We recast the problem as a hybrid in which a partial, identifier-tracked micro-sample supplies the deterrence (cost) structure of an entropic optimal-transport / gravity model whose margins are pinned by the station counts. Two results follow. First, the transport cost is identifiable only modulo additive station effects; the bias induced by incomplete observation equals the *non-separable* component of the log-selection probability, so station-emptiness censoring cancels exactly, whereas 60-second polling—whose capture probability is a function of travel time, the very quantity being estimated—does not. Second, the estimator yields a *sampling-horizon law*: the continuous-collection time needed to reconstruct OD to target accuracy δ scales as δ^{-4} and inversely with the identifier-persistence rate q . The law implies a standard-compliance threshold below which operational OD monitoring is infeasible. We give a feed-by-feed audit protocol instantiating the bound across [To confirm: N] French networks. [To confirm: insert headline numbers from the pilot once collected.]

1 Introduction

[To confirm: Three paragraphs. (1) OD flows are the primitive of bike-sharing planning, rebalancing and equity analysis, yet public GBFS feeds were designed to advertise availability, not to record trips; the literature is cleaved into underdetermined count-based inference and identifier-based tracking, each with a distinct failure mode. (2) Our reframe: feasibility of reconstruction is not a modelling choice but a function of how far a feed complies with the GBFS identifier-rotation guidance and of the polling cadence; we make this quantitative. (3) Contributions and roadmap: an identifiability characterisation (Sec. 3), a selection-bias decomposition isolating aliasing as the dominant threat (Sec. 4), a sampling-horizon law and a standard-compliance threshold (Sec. 5), and a feed-by-feed audit protocol with a pilot (Sec. 6).]

2 Data regimes, model and notation

[To confirm: Prose introducing the three regimes (published trips; station-status counts; persistent-identifier tracking) and positioning our hybrid between the latter two.]

Let stations be indexed $i, j \in \{1, \dots, N\}$ and let $P^* \in \mathbb{R}_{\geq 0}^{N \times N}$ be the true OD coupling over a fixed period, with row margins $r_i^* = \sum_j P_{ij}^*$ (outflow) and column margins $c_j^* = \sum_i P_{ij}^*$ (inflow). The structural (gravity / entropic optimal-transport) assumption is the Gibbs form

$$P_{ij}^* = u_i^* v_j^* e^{-c_{ij}^*/\varepsilon}, \quad (1)$$

for potentials u^*, v^* , ground cost c^* and temperature $\varepsilon > 0$. This is the maximum-entropy trip-distribution model (Wilson, 1967) and is identical, as an algorithm, to iterative proportional

fitting and to the Sinkhorn iteration for entropic optimal transport (Sinkhorn, 1964; Cuturi, 2013; Peyré and Cuturi, 2019).

Macro observation (censored margins). Station-status counts provide noisy, right-censored margins $\hat{r}_i = \rho_i r_i^*$ with availability fraction $\rho_i \in (0, 1]$, reflecting demand lost at empty/full stations (Rudloff and Lackner, 2014; Negahban, 2019).

Micro observation (selection-tilted sample). A sub-sample of identifier-tracked trips is drawn from the *selection-tilted* law $\tilde{P}_{ij} \propto P_{ij}^* S_{ij}$, where $S_{ij} \in [0, 1]$ aggregates identifier persistence q , non-rebalancing $(1 - \beta)$, and capture under polling interval Δ (anti-aliasing) κ .

3 Identifiability of the interaction cost

The Gibbs form (1) is invariant under $c_{ij}^* \mapsto c_{ij}^* + f_i + g_j$ (absorbed into the potentials). Hence the cost is identifiable only modulo the separable subspace $\mathcal{N} = \{(f_i + g_j)_{ij}\}$ ($\dim \mathcal{N} = 2N - 1$); the identifiable object is the interaction part $\Pi_{\mathcal{N}^\perp} c^*$, equivalently the second-order cross-differences $c_{ij} - c_{il} - c_{kj} + c_{kl}$.

Theorem 1 (Cost identifiability). *Under (1), $\Pi_{\mathcal{N}^\perp} c^*$ is identifiable from a coupling, and no separable component of c^* is. [To confirm: State regularity conditions; proof by the matching/ground-metric argument of Galichon and Salanié (2022); Dupuy and Galichon (2014); Cuturi and Avis (2014); full proof in SI Appendix A.]*

Definition 2 (Admissible cost feature). A candidate feature $\psi_{ij} = \phi^{(k)}(x_i, x_j)$ is *admissible* iff it survives two-way (origin/destination) centring, $\tilde{\psi}_{ij} = \psi_{ij} - \bar{\psi}_{i\cdot} - \bar{\psi}_{\cdot j} + \bar{\psi}_{\cdot\cdot} \neq 0$; monadic features ($\psi_{ij} = a_i$ or b_j) are inadmissible and belong in the potentials/margins, not the cost.

[To confirm: Discuss the low-rank parametric cost $c_{ij} = \phi(x_i, x_j; \theta)$: **admissible, mutually non-collinear-after-projection dyadic features** — **routed cycling distance, directed elevation gain, circuitry, protected-lane fraction, barrier (river/rail) crossings**. Conditioning $\lambda_{\min}$ of the projected information matrix is what keeps the collection horizon finite (Sec. 5).]

4 The selection-bias decomposition

Proposition 3 (Bias = non-separable log-selection). *The micro-sample identifies the cost of $\tilde{P}$, namely $\tilde{c} = c^* - \varepsilon \log S$; therefore the estimated identifiable cost satisfies*

$$\Pi_{\mathcal{N}^\perp} \hat{c} \xrightarrow{n \rightarrow \infty} \Pi_{\mathcal{N}^\perp} c^* - \varepsilon \Pi_{\mathcal{N}^\perp} \log S. \quad (2)$$

In particular, if the selection is separable, $S_{ij} = a_i b_j$, then $\log S \in \mathcal{N}$ and the bias vanishes.

Corollary 4 (Which censoring hurts). *Station-emptiness censoring acts at the origin/destination level ($S_{ij} = a_i b_j$) and cancels in the identifiable cost. Aliasing under polling interval Δ has capture probability that depends on the trip duration t_{ij} , hence on c_{ij}^* itself; it is non-separable and correlated with the estimand, so it does not cancel and systematically attenuates the cost–distance relationship. Residual operator rebalancing on specific corridors is likewise non-separable.*

[To confirm: Quantify the residual non-separable bias as a function of Δ ; proof of Prop. 3 in SI Appendix A.]

5 The sampling-horizon law

Writing Λ for the trip-generation rate (trips/day), the number of clean, attributable observations after T days of polling at interval Δ is

$$n_{\text{eff}} = \Lambda T \kappa(\Delta) q (1 - \beta) (1 - \gamma), \quad (3)$$

with γ the GPS-jitter / collision loss. Propagating the cost error through the entropic-OT forward map (sensitivity $O(1/\varepsilon)$, entropic bias $O(\varepsilon)$; balancing gives an $n^{-1/4}$ rate on the OD error) yields the requirement $n_{\text{eff}} \gtrsim \dim \theta / (\lambda_{\min} \delta_{\text{OD}}^4)$ and hence the collection horizon.

Theorem 5 (Sampling-horizon law). *To reconstruct the OD coupling to relative accuracy δ_{OD} under the estimator of Sections 3–4,*

$$T^* = \frac{\dim \theta}{\lambda_{\min} \delta_{\text{OD}}^4 \Lambda \kappa(\Delta) q (1 - \beta) (1 - \gamma)}, \quad (4)$$

so $T^* \propto \delta_{\text{OD}}^{-4} q^{-1} \Lambda^{-1} \lambda_{\min}^{-1}$. [To confirm: State constants and the forward-stability lemma; proof in SI Appendix B.]

Corollary 6 (Standard-compliance threshold). *For any operational horizon $T_{\max}$ and accuracy δ_{OD} , OD is reconstructible only on feeds with identifier persistence $q \geq q_{\min} := \dim \theta / (\lambda_{\min} \delta_{\text{OD}}^4 \Lambda \kappa (1 - \beta)(1 - \gamma) T_{\max})$. Below $q_{\min}$ the horizon exceeds $T_{\max}$ and operational monitoring is infeasible.*

[To confirm: Pooling rescue: a hierarchical cost shared across the N networks divides $\dim \theta$ per city and buys down T^* ; this is why the multi-system panel is statistically load-bearing, not merely comparative. Also state the comparability tension: a fixed ε across cities imposes a precision floor $\delta_{\text{OD}} \gtrsim \varepsilon$.]

6 Empirical protocol and pilot

[To confirm: The five constants $(q, \kappa, \beta, \gamma, \Lambda)$ are unknown and feed-specific; the protocol is a 7-day pilot that *measures* them per feed (script experiments/d01_pilot.py). $\hat{q}$: fraction of identifiers persisting origin-to-destination. $\hat{\kappa}(\Delta)$: by re-reconstructing at coarser intervals and measuring trip loss (the aliasing curve). $\hat{\beta}$: is_disabled + departure-burst fraction. $\hat{\Lambda}$: reconstructed trips/day. $\lambda_{\min}$: smallest eigenvalue of the projected dyadic-feature Gram matrix. Instantiate (4) per network; produce the thesis figure T^* vs q with the operational-monitoring frontier and the [To confirm: N] French feeds positioned on it.]

7 Related work

[To confirm: Position at the empty intersection of (i) count-based OD estimation (Van Zuylen and Willumsen, 1980; Cascetta, 1984; Cascetta and Nguyen, 1988; Spiess, 1987; Bell, 1991; Ma and Qian, 2023), (ii) identifier-based micromobility OD inference (Xu et al., 2022), (iii) censored-demand correction (Rudloff and Lackner, 2014; Negahban, 2019; Goh et al., 2019), (iv) inverse / ground-metric optimal transport (Cuturi and Avis, 2014; Stuart and Wolfram, 2020; Dupuy and Galichon, 2014), and (v) feed quality / standards auditing (our GBFS Audit Catalogue, Paper 01). One sentence of "what we add" per cluster. Note the equivalence IPF = Sinkhorn = gravity to pre-empt a novelty objection, and that network tomography (Vardi, 1996) loses its identifying routing matrix in the station-count setting.]

8 Discussion

[To confirm: Standards reading: the GBFS identifier-rotation guidance (issue #146; v2.0) is a privacy feature that simultaneously sets the ceiling on OD observability; Corollary 6 turns this into a quantitative recommendation (a minimum persistence, or an explicit acknowledgement that OD is non-monitorable from compliant feeds). Connect to MDS and to the regulator-facing data path.]

9 Limitations and future work

[To confirm: The Gibbs/low-rank-cost assumption is the binding modelling commitment; the bound is conditional on it. Constants in Theorem 5 from entropic-OT stability can be loose at $N \sim 1500$ (Mena and Niles-Weed, 2019; Genevay et al., 2019). The estimator recovers the entropy-optimal representative, not pair-level OD; interior cells remain weakly identified. Validation against ground truth requires a feed that both exposes persistent identifiers and publishes trips—address head-on.]

10 Methods

[To confirm: Reproducibility-grade Methods: feed catalogue and discovery, the vehicle snapshot collector (60 s), trip reconstruction (min-move, rebalancing heuristics), the cost-feature pipeline (routing, elevation, infrastructure layers), the inverse-OT cost estimator, the forward Sinkhorn solver, and the constant-estimation procedures. Software stack with version pins; SEED = 42.]

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All data sources and per-experiment derived outputs are released under the MIT licence at <https://github.com/cycling-data-lab/gbfs-od-reconstruction>. A frozen Zenodo snapshot of the v1.0 release tag is deposited at <https://doi.org/10.5281/zenodo.XXXXXXX> **[To confirm: substitute the issued Zenodo DOI after the first release]**.

Reproducibility

Each table and figure is produced by a numbered script `experiments/d<NN>_<name>.py` that consumes a known input cache, writes a named output JSON, and pins SEED=42. **[To confirm: Add experiment-specific reproducibility detail here.]**

Competing interests

The authors declare no competing financial or non-financial interests.

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